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$$f(x) = \frac{\sqrt{x} - 1}{\sqrt[3]{x} - 1}$$

$$\lim_{x \rightarrow 1} f(x) = \frac{1 - 1}{1 - 1} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{(\sqrt{x} - 1)(\sqrt{x} + 1) \left( (\sqrt[3]{x})^2 + \sqrt[3]{x} + 1 \right)}{(\sqrt[3]{x} - 1)(\sqrt{x} + 1) \left( (\sqrt[3]{x})^2 + \sqrt[3]{x} + 1 \right)} = \lim_{x \rightarrow 1} \frac{\cancel{(x-1)} \left( (\sqrt[3]{x})^2 + \sqrt[3]{x} + 1 \right)}{\cancel{(x-1)}(\sqrt{x} + 1)} =$$

$$\frac{1 + 1 + 1}{1 + 1} = \frac{3}{2}$$

## References